

Mara Schwartz

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EXPERIENCE

NORTH ATLANTIC INDUSTRIES | BOHEMIA, NY

Software Engineer II | September 2025 - Present

- Leading PinMapper into a scalable, full-stack solution deployed in a publicly-released product for internal teams and customers
- Own end-to-end development across frontend (React), backend APIs, CI/CD pipelines, and AWS cloud infrastructure (ECS, Fargate)
- Leading development of an internal AI Agent for engineering, leveraging a hybrid Vector RAG and Graph RAG approach for context delivery
- Manage one Junior SWE and six interns, providing technical mentorship and project oversight

Software Engineer I | July 2024 - September 2025

- Led PinMapper, advancing a decade-old project into a production-ready product; cut pin-out creation time from 4+ hours to under 1 minute
- Developed automated thermal testing tools for COTS hardware; surfaced critical issues in pre-production, preventing contract risk
- Delivered critical documentation to COTS customers, test engineers, and application engineers
- Implemented two-week sprint cadence and Jira workflow for a 10-member team, increasing planning visibility and on-time delivery
- Designed and delivered product demos, earning company recognition for technical communication and customer engagement

LANGUAGE AND COGNITION ARTIFICIAL INTELLIGENCE LAB (LACAI LAB) | RENSSELAER POLYTECHNIC INSTITUTE

Graduate Research Assistant | August 2022 - February 2023 | Troy, NY

- Researcher on GATOR, an IBM-funded research project to prototype a new task-oriented human-machine dialogue system using transformers
- Co-authored and published our results in the AHFE International 2023 conference proceedings

Undergraduate Research Assistant | May 2020 - August 2021 | Troy, NY

- Lead developer for the Socio-Cultural Content in Language (SCIL) toolset, an NLP toolset for identifying social roles within dialogues

PROJECTS

MASTER'S PROJECT | PYTHON, PYTORCH, TRANSFORMERS, SCIKIT-LEARN, NUMPY

Principal Investigator | 2022

- Performed dialogue analysis on Twitter user clusters to determine how influence campaigns propagate influence through social networks
- Identified prominent influencers and influential opinions within the user clusters using the SCIL toolset on constructed dialogues

SOCIO-CULTURAL CONTENT IN LANGUAGE (SCIL) | PYTHON, STANFORD CORENLP, NLTK, PYTORCH, TRANSFORMERS, SCIKIT-LEARN, KERAS

Project Lead, Developer | 2020-2021

- Developed an improved NLP toolset for identifying social roles in dialogues, reducing processing time from about 5 minutes to 5 seconds
- Fine-tuned a 12-layer BERT model for automatic dialogue act classification and tagging

YET ANOTHER COURSE SCHEDULER (YACS) | ANGULAR, RUBY ON RAILS, TYPESCRIPT, POSTGRESQL, DOCKER

Project Lead, Developer, Mentor | 2018 - 2019

- Led and mentored a 12-person team in the Rensselaer Center for Open Source (RCOS) to improve YACS, a tool used by over 5,000 students
- Contributed to open-source development by enhancing functionality with multi-semester planning and admin tools

EDUCATION

MASTER OF SCIENCE IN COMPUTER SCIENCE | RENSSELAER POLYTECHNIC INSTITUTE

August 2021 - December 2022

GPA: 3.92 | FOCUS: MACHINE LEARNING AND NATURAL LANGUAGE PROCESSING

- Coursework: Machine Learning & Optimization, Distributed Systems & Algorithms, Parallel Computing, Computing & Quantum Computing
- Teaching: Teaching Assistant (Programming Languages, Data Structures); designed assignments, mentored students, led labs

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & COGNITIVE SCIENCE | RENSSELAER POLYTECHNIC INSTITUTE

August 2017 - May 2021

GPA: 3.86 | MAGNA CUM LAUDE | MINOR IN MATHEMATICS

- Coursework: Social Computing (NLP), Intro to AI, Cognitive Modeling, Graph Theory, Numerical Computing
- Honors: UPE Honor Society, Rensselaer Leadership Award, Dean's Honor List
- Teaching: Teaching Assistant (Operating Systems); Computer Science Tutor (Intro CS, Data Structures)

PUBLICATIONS

[1] A. Sanders, M. Schwartz, A. Chang, S. Briggs, J. Braasch, D. Wang, M. Si, and T. Strzalkowski, Towards a proper evaluation of automated conversational systems, in Artificial Intelligence and Social Computing, vol. 72, July 2023. doi: 10.54941/ahfe1003276.

SKILLS

- **Programming Languages:** C/C++, Python, JavaScript/Typescript, SQL, C#, Java, Erlang
- **Libraries/Frameworks:** PyTorch, Transformers, TensorFlow, HuggingFace, scikit-learn, Pandas, OpenCL, OpenCV
- **Web Technologies:** React, Vite, Redux, TanStack Query, Tailwind CSS, Next.js, Django, Angular
- **Infrastructure & DevOps:** Docker, AWS (ECS, EC2, Fargate), Jenkins, GitLab CI/CD
- **Tools:** Git, Linux/Unix, CMake, Bash, Jira, Agile/Scrum, Jupyter, L^AT_EX